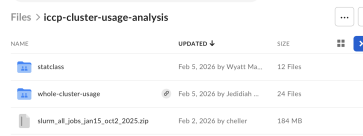


Date Preprocessing

Jiachen, Yanlin

Data Source Selection and Its Relationship to Preprocessing

The raw data are distributed across multiple folders, each corresponding to a different data-generating pipeline or intended use case. In our project, the goal is to study *job submission behavior* and *execution scheduling outcomes* at the cluster-wide level. Therefore, we select the data source that (i) covers the full population of jobs on the cluster, (ii) contains authoritative scheduling timestamps, and (iii) provides a consistent schema suitable for building job-level analytical tables.

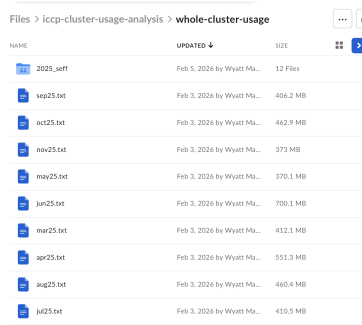


NAME	UPDATED	SIZE
statclass	Feb 5, 2026 by Wyatt Ma...	12 Files
whole-cluster-usage	Feb 5, 2026 by Jaidiah ...	24 Files
slurm_all_jobs_jan15_oct2_2025.zip	Feb 5, 2026 by cheller	194 MB

Figure 1: The Overview of the Data & Folders

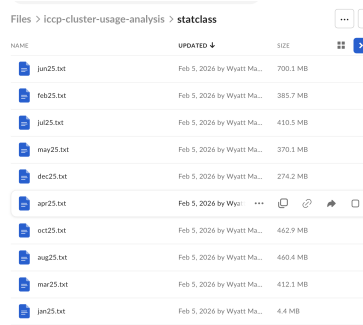
We **do not use** the preliminary 2.7GB `slurm_all_jobs_jan15_oct2_2025.zip` file because it represents a partial-year extract with an incomplete temporal coverage. Since our analysis focuses on submission timing patterns and execution schedules over a consistent annual horizon, using a truncated time window would introduce seasonality bias and distort temporal comparisons. Therefore, we rely exclusively on the full-year 2025 accounting exports.

Folder roles (what each folder represents). Conceptually, the directory contains the following types of data:



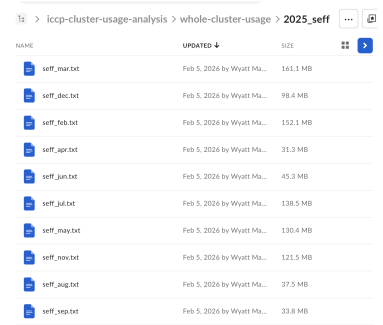
NAME	UPDATED	SIZE
2025_seff	Feb 5, 2026 by Wyatt Ma...	12 Files
sep25.txt	Feb 5, 2026 by Wyatt Ma...	406.2 MB
oct25.txt	Feb 5, 2026 by Wyatt Ma...	462.9 MB
nov25.txt	Feb 5, 2026 by Wyatt Ma...	373 MB
may25.txt	Feb 5, 2026 by Wyatt Ma...	376.1 MB
jun25.txt	Feb 5, 2026 by Wyatt Ma...	700.1 MB
mar25.txt	Feb 5, 2026 by Wyatt Ma...	412.1 MB
apr25.txt	Feb 5, 2026 by Wyatt Ma...	551.3 MB
aug25.txt	Feb 5, 2026 by Wyatt Ma...	460.4 MB
jul25.txt	Feb 5, 2026 by Wyatt Ma...	410.5 MB

(a) whole-cluster-usage/



NAME	UPDATED	SIZE
jan25.txt	Feb 5, 2026 by Wyatt Ma...	700.1 MB
feb25.txt	Feb 5, 2026 by Wyatt Ma...	385.7 MB
jul25.txt	Feb 5, 2026 by Wyatt Ma...	410.5 MB
may25.txt	Feb 5, 2026 by Wyatt Ma...	370.1 MB
dec25.txt	Feb 5, 2026 by Wyatt Ma...	274.2 MB
apr25.txt	Feb 5, 2026 by Wyat...	...
oct25.txt	Feb 5, 2026 by Wyatt Ma...	462.9 MB
aug25.txt	Feb 5, 2026 by Wyatt Ma...	460.4 MB
mar25.txt	Feb 5, 2026 by Wyatt Ma...	412.1 MB
jan25.txt	Feb 5, 2026 by Wyatt Ma...	4.4 MB

(b) statclass/



NAME	UPDATED	SIZE
seff_mar.txt	Feb 5, 2026 by Wyatt Ma...	141.1 MB
seff_dec.txt	Feb 5, 2026 by Wyatt Ma...	98.4 MB
seff_feb.txt	Feb 5, 2026 by Wyatt Ma...	152.1 MB
seff_apr.txt	Feb 5, 2026 by Wyatt Ma...	31.3 MB
seff_jun.txt	Feb 5, 2026 by Wyatt Ma...	45.3 MB
seff_jul.txt	Feb 5, 2026 by Wyatt Ma...	138.5 MB
seff_may.txt	Feb 5, 2026 by Wyatt Ma...	130.4 MB
seff_nov.txt	Feb 5, 2026 by Wyatt Ma...	121.5 MB
seff_aug.txt	Feb 5, 2026 by Wyatt Ma...	37.5 MB
seff_sep.txt	Feb 5, 2026 by Wyatt Ma...	33.8 MB

(c) 2025_seff/

Figure 2: Directory structure overview and folder roles.

1. **whole-cluster-usage/ (production cluster logs):** cluster-wide records for all users and partitions, intended for monitoring and accounting. **This is the only source that represents the full target population for our analysis.**

2. **statclass/ (teaching/sample data):** a small, course-oriented subset used for instructional purposes rather than population-level inference. Because it is not representative of overall cluster usage, we **do not** mix it with production logs.
3. **Auxiliary subfolders within whole-cluster-usage/2025_seff/:** post-processed summaries focused on efficiency metrics (CPU/memory utilization) and may be incomplete (e.g., only available for a subset of jobs or only after completion). These are treated as optional secondary sources and are **not used** in the core construction pipeline to avoid mixing raw logs with derived reports.

In summary, our analysis is conducted using the accounting logs under **whole-cluster-usage/**, while excluding auxiliary subfolders such as **2025_seff/**, which contain derived efficiency summaries rather than primary scheduling records.

Dataset Construction and Processing Pipeline

Data Source and Directory Structure

The 2025 accounting data are provided as twelve monthly, pipe-delimited text files (**jan25.txt** through **dec25.txt**), with a consistent schema of 120 columns. These columns include identifiers (e.g., **JobIDRaw**, **User**, **Account**), scheduling metadata (**Partition**, **QOS**), timestamps (**Submit**, **Start**, **End**), and resource request/allocation variables.

The monthly files represent time partitions of the same logging system. To construct the full 2025 dataset, we vertically **concatenate the twelve accounting tables into a single merged input** named **whole_cluster_usage_merged_raw.txt.gz**.

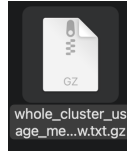


Figure 3: whole_cluster_usage_merged_raw.txt.gz

Raw Input and Parsing

The merged accounting log contains millions of snapshot-level records. Because of its size (approximately 5GB), we process the file in chunks to avoid memory overflow.

During ingestion, we apply minimal normalization:

- **Missing-value normalization:** we treat tokens such as "", **unknown**, **none**, and **n/a** (case-insensitive) as missing values.
- **Cohort restriction (Submit year):** we restrict to jobs whose **Submit** timestamp begins with “2025”, defining the 2025 submission cohort.
- **State simplification:** **State** is reduced to its first token (e.g., “CANCELLED by ...” → “CANCELLED”) to consolidate state categories.

File 1: master_2025_raw_valid.csv.gz (Row-level snapshots)

We first build a row-level “raw but usable” dataset for year 2025, denoted **master_2025_raw_valid.csv.gz**. This table retains only 28 of analysis-relevant columns and is written incrementally (chunk-by-chunk) to a compressed CSV.

3 Derived variables and unit harmonization. To make resource and timing variables comparable across jobs, we derive:

- **Requested memory in MB:** parse ReqMem strings (e.g., 500M, 2G, 1.5T) and convert to a numeric ReqMem_MB.
- **Wait and run times:** compute

$$\text{WaitTimeSec} = \text{Start} - \text{Submit}, \quad \text{RunTimeSec} = \text{End} - \text{Start},$$

in seconds when timestamps are available.

Construct Job-level Tables

The raw SLURM log can contain **multiple snapshot records** for the same JobIDRaw (e.g., a job may appear as pending, then running, then completed). For modeling and interpretability, we construct job-level tables that contain exactly one row per JobIDRaw, with different selection rules depending on the analysis goal.

To avoid loading the full dataset into memory for global de-duplication, we use a hash-based bucketing strategy: JobIDRaw values are hashed into 256 buckets, written to temporary files, and de-duplicated within each bucket. This enables scalable job-level selection on a laptop.

File 2: master_2025_joblevel_submission.csv.gz (Submission-view)

The submission-view table contains one row per job, representing the **earliest submission snapshot**. Within each bucket, we sort rows by JobIDRaw, then by submit time (and original row order as a tie-breaker), and keep the first record for each JobIDRaw. This table supports “job submission behavior” analyses, such as submission timing patterns and requested resource profiles.

File 3: master_2025_joblevel_terminal.csv.gz (Terminal/execution-view)

The terminal-view table also contains one row per job, but it aims to represent the *final or most complete* snapshot for execution and outcome analysis. Within each bucket, we prioritize records as follows:

1. records with a valid End timestamp,
2. otherwise records with a valid Start timestamp,
3. otherwise the submission-only record.

Among records with the same priority, we prefer the latest available time key (end/start/submit) to obtain the most up-to-date state. This table supports “execution schedules” analyses including wait time, runtime, and outcome rates (e.g., FAILED, TIMEOUT, OUT_OF_MEMORY).

Schema Summary

Column	Raw Valid	Submission	Terminal
JobIDRaw	Yes (Original)	Yes (Original)	Yes (Original)
JobID	Yes (Original)	–	–
JobName	Yes (Original)	–	–
User	Yes (Original)	Yes (Original)	Yes (Original)
Group	Yes (Original)	Yes (Original)	Yes (Original)
Account	Yes (Original)	Yes (Original)	Yes (Original)
Partition	Yes (Original)	Yes (Original)	Yes (Original)
QOS	Yes (Original)	Yes (Original)	Yes (Original)

Column	Raw Valid	Submission	Terminal
Submit	Yes (Original)	Yes (Original)	Yes (Original)
Eligible	Yes (Original)	–	Yes (Original)
Priority	Yes (Original)	Yes (Original)	Yes (Original)
Start	Yes (Original)	–	Yes (Original)
End	Yes (Original)	–	Yes (Original)
State	Yes (Original)	Yes (Original)	Yes (Original)
Reason	Yes (Original)	–	Yes (Original)
ExitCode	Yes (Original)	–	Yes (Original)
ReqCPUS	Yes (Original)	Yes (Original)	Yes (Original)
ReqMem	Yes (Original)	–	–
ReqMem_MB	Yes (Derived)	Yes (Derived)	Yes (Derived)
ReqNodes	Yes (Original)	Yes (Original)	Yes (Original)
ReqTRES	Yes (Original)	Yes (Original)	Yes (Original)
AllocCPUS	Yes (Original)	Yes (Original)	Yes (Original)
AllocNodes	Yes (Original)	Yes (Original)	Yes (Original)
AllocTRES	Yes (Original)	–	–
ElapsedRaw	Yes (Original)	–	Yes (Original)
TimelimitRaw	Yes (Original)	Yes (Original)	Yes (Original)
WaitTimeSec	Yes (Derived)	–	Yes (Derived)
RunTimeSec	Yes (Derived)	–	Yes (Derived)

Column Counts

- master_2025_raw_valid.csv.gz: 28 columns
- master_2025_joblevel_submission.csv.gz: 16 columns
- master_2025_joblevel_terminal.csv.gz: 24 columns